Neural Retail

Data Engineering & ML Project Report

Retail Analytics Dashboard with ETL Pipeline,
SQL Analysis, and Machine Learning Forecasting

Project Type	End-to-End Data Engineering & Analytics
Domain	Neura-Retail :-AI Powered Sales Intelligence System
Technologies	Python, PostgreSQL, Streamlit, Jupyter notebook , PowerBi
Date	26 April 2026

1. Project Objective

This project aims to build a complete end-to-end data engineering and analytics system for retail sales data. The system processes raw data, stores it in a relational database, performs business-level SQL analysis, and applies machine learning models for forecasting and churn prediction.

2. Data Collection

Dataset: Retail transaction dataset (CSV format)

The dataset contains the following fields:

- Invoice ID – Unique transaction identifier
- StockCode & Description – Product reference and name
- Quantity – Number of units purchased
- InvoiceDate – Date and time of transaction
- Price – Unit price of the product
- Customer ID – Unique customer identifier
- Country – Country of transaction origin

3. Data Cleaning & Preprocessing

Performed using Python (Pandas):

- Removed rows with missing Customer ID or Date values
- Fixed date format inconsistencies in the InvoiceDate column
- Created derived column: TotalPrice = Quantity x Price
- Resolved inconsistent formatting throughout the dataset
- Ensured clean, structured data for downstream processing

4. ETL Pipeline Development

A dedicated Python script (etl_pipeline.py) was developed to manage the full Extract–Transform–Load workflow:

4.1 Extract

- Reads raw CSV data into a Pandas DataFrame

4.2 Transform

- Applies all cleaning and preprocessing steps described in Section 3

4.3 Load

- Inserts the cleaned data into a PostgreSQL table (retail_data) within the retail_db database

5. Database Integration (PostgreSQL)

- Connected Python to PostgreSQL using the psycopg2 library
- Created the database: retail_db
- Defined and populated the retail_data table
- Verified data integrity using SQL queries via pgAdmin

6. SQL Analysis & Business Insights

The following business questions were answered through structured SQL queries:

Category	Tools / Libraries
Total Revenue	Aggregate revenue across all transactions
Total Customers	Count of unique customer IDs
Total Orders	Count of unique invoice IDs
Top 10 Products	Ranked by total quantity sold
Top 10 Customers	Ranked by total revenue generated
Revenue by Country	Breakdown of sales per country
Sales Trend	Revenue aggregated over time

7. Streamlit Dashboard

An interactive dashboard (app.py) was built using Streamlit to visualize key metrics and trends.

7.1 KPI Metrics

- Total Revenue
- Total Customers
- Total Orders

7.2 Visualizations

- Top Products by Quantity
- Top Customers by Revenue

- Revenue Distribution by Country
- Sales Trend over Time

7.3 Filters

- Country selection filter for dynamic data slicing

8. Churn Prediction Model

File: churn_model.py

8.1 Feature Engineering

- RFM (Recency, Frequency, Monetary) features derived from transaction history
- Churn defined as: customers inactive for more than 90 days

8.2 Model

- Algorithm: Logistic Regression
- Outputs: Accuracy score and full classification report

9. Sales Forecasting Model

File: forecast_model.py

9.1 Approach

- Aggregated transaction data at the daily level
- Applied Linear Regression for short-term prediction

9.2 Outputs

- Predicted sales for the next 30 days
- Export: sales_forecast_30_days.csv
- Visualization: sales_forecast_chart.png (Actual vs Predicted)

9.3 Dashboard Integration

- Forecast table and Actual vs Predicted chart added to the Streamlit dashboard

10. Technologies Used

Category	Tools / Libraries
Language	Python 3.x
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn (Logistic Regression, Linear Regression)
Database	PostgreSQL (pgAdmin)
Query Language	SQL
Dashboard	Streamlit
Visualization	Matplotlib
DB Connector	psycopg2

11. Key Learnings

- End-to-end ETL pipeline design and implementation
- Python-to-PostgreSQL integration using psycopg2
- SQL-based business analytics and reporting
- Interactive dashboard creation with Streamlit
- Supervised machine learning model development
- Fundamentals of time-series sales forecasting

12. Future Enhancements

- Implement advanced forecasting models: Facebook Prophet, LSTM (deep learning)
- Integrate gradient boosting models: XGBoost, LightGBM for churn prediction
- Redesign and improve the dashboard UI/UX
- Deploy the project online (cloud hosting / containerisation)

13. Conclusion

This project successfully demonstrates a complete, production-style data engineering, analytics, and machine learning pipeline. Starting from raw retail transaction data, the system delivers cleaned and structured storage, rich SQL-driven business insights, an interactive dashboard, and predictive capabilities through churn and sales forecasting models. The project serves as a strong foundation for further enhancements and real-world deployment.